

$d = 4, k = 2$ (size 15) $M_k(y)_{\alpha, \beta} = y_{\alpha + \beta}$

deg 0 deg 1

deg 2

deg 0	$y_{ \alpha =0}$	$y_{ \alpha =1}$	$y_{ \alpha =2}$
deg 1	$y_{ \alpha =1}$	$y_{ \alpha =2}$	$y_{ \alpha =3}$
deg 2	$y_{ \alpha =2}$	$y_{ \alpha =3}$	$y_{ \alpha =4}$

Lasserre conditions

$$y_0 = 1$$

normalization

$$y_\alpha \geq 0$$

$x \geq 0$ on Δ_d

$$y_\alpha = \sum_{\sum_i x_i = 1} y_{\alpha + e_i}$$

$$M_k(y) \geq 0$$

moment PSD

$$M_{k-1}(x_i \cdot y) \geq 0$$

localizing PSD

$$t \cdot M_{k-1}(y) - \sum M_W[i, j] M_{k-1}(x_i x_j \cdot y) \geq 0$$

window localizing